01 - Computational time analysis

September 29, 2026

1 Introduction

This document briefly presents the results of the computational time analysis on the python code `do_features.py` which computes the audio features of the audio files from the FMA dataset. These audio features will be used in order to perform a classification by genre.

2 Results

The analysis has been made on the first 200 audio files from the `FMA_small` dataset [1]. The results are summed up in the table 1 and table 2, and the first raws are shown on figure 1 and figure 2.

The computational time is dominated by the computation of the HPSS feature (stands for Harmonic Percussive Source Separation). It represents around two thirds of the computational time. So this feature needs to be investigated in details. If this feature is not very useful and does not improve geniunely the performances of the classifier, it will most likely be removed from the feature computation.

Appart from the HPSS feature, the computational time of the audio features are of the same order of magnitude. The second most expensive computation is `cqt_compute`. It is used to compute 4 different audio features: `chroma_cqt`, `chroma_cens`, `tonnetz_cqt` and `tonnetz_cens`. If these features are not relevant enough for the classification algorithms, they can be removed from the feature computation to save some time.

In the end, all features will be investigated, and the ones that are not relevant enough for the classification algorithm will be removed from the computation. This will come in another analysis.

Table 1: The results of the computational time analysis for all the audio features. The numbers give the mean, standard deviation, minimum and maximum of the computational time for the full computation of all features, called `total`, and then for the computation of each feature, called `feature_compute`, and the computation of the statistics (mean, standard deviation, skew, kurtosis, median, min, max, 10th and 90th percentils) of each feature, called `feature_stats`. The table has been truncated. The following lines are presented in table 2.

	mean	std	min	max
total	3.408	0.123	3.254	4.755
hpss_compute	2.343	0.053	2.245	2.601
cqt_compute	0.181	0.009	0.170	0.282
beat_tracking_compute	0.090	0.059	0.078	0.916
chroma_stft_compute	0.084	0.005	0.077	0.107
load_audio	0.070	0.021	0.028	0.300
spectral_bandwidth_compute	0.055	0.006	0.048	0.086
onset_strength_compute	0.054	0.011	0.044	0.095
spectral_rolloff_compute	0.053	0.002	0.050	0.061
tempogram_compute	0.043	0.010	0.032	0.081
zcr_compute	0.035	0.008	0.034	0.141
stft_compute	0.035	0.003	0.029	0.056
spectral_rolloff_stats	0.026	0.003	0.024	0.046
spectral_contrast_compute	0.022	0.001	0.021	0.029
spectral_centroid_compute	0.021	0.001	0.020	0.027
harmonic_percussive_rms_stats	0.017	0.002	0.016	0.033
harmonic_percussive_ratio_stats	0.017	0.002	0.016	0.027
mel_compute	0.014	0.011	0.004	0.044
chroma_stft_stats	0.013	0.003	0.010	0.029
mfcc_stats	0.013	0.003	0.012	0.035
delta_mfcc_stats	0.013	0.002	0.012	0.033
delta2_mfcc_stats	0.013	0.001	0.012	0.025
chroma_cqt_stats	0.012	0.001	0.010	0.017
delta_chroma_stft_stats	0.011	0.002	0.009	0.028
spectral_flatness_compute	0.011	0.001	0.011	0.016
tonnetz_cqt_stats	0.011	0.001	0.010	0.015
tonnetz_cens_stats	0.011	0.001	0.009	0.017
delta_chroma_cqt_stats	0.011	0.001	0.009	0.017
chroma_cens_stats	0.011	0.001	0.009	0.015

Table 2: The following lines of the table 1.

	mean	std	min	max
onset_strength_stats	0.010	0.002	0.009	0.025
spectral_contrast_stats	0.010	0.001	0.009	0.020
tempogram_stats	0.010	0.002	0.008	0.024
rms_stats	0.010	0.001	0.009	0.023
zcr_stats	0.009	0.001	0.008	0.012
spectral_bandwidth_stats	0.009	0.001	0.008	0.014
spectral_centroid_stats	0.009	0.001	0.008	0.013
spectral_flatness_stats	0.009	0.001	0.008	0.012
spectral_flux_stats	0.009	0.001	0.008	0.015
beat_interval_stats	0.008	0.001	0.008	0.013
mfcc_compute	0.005	0.001	0.005	0.012
spectral_flux_compute	0.004	0.001	0.003	0.007
rms_compute	0.003	0.001	0.002	0.009
chroma_cens_compute	0.002	0.000	0.002	0.005
harmonic_percussive_rms_compute	0.002	0.000	0.001	0.003
stft_power	0.001	0.000	0.001	0.002
harmonic_percussive_ratio_compute	0.001	0.000	0.001	0.002
delta2_mfcc_compute	0.001	0.000	0.001	0.004
chroma_cqt_compute	0.001	0.001	0.001	0.004
delta_mfcc_compute	0.001	0.000	0.001	0.004
delta_chroma_stft_compute	0.001	0.000	0.001	0.003
delta_chroma_cqt_compute	0.001	0.000	0.001	0.002
tonnetz_cqt_compute	0.000	0.000	0.000	0.000
tonnetz_cens_compute	0.000	0.000	0.000	0.000
beat_interval_compute	0.000	0.000	0.000	0.000

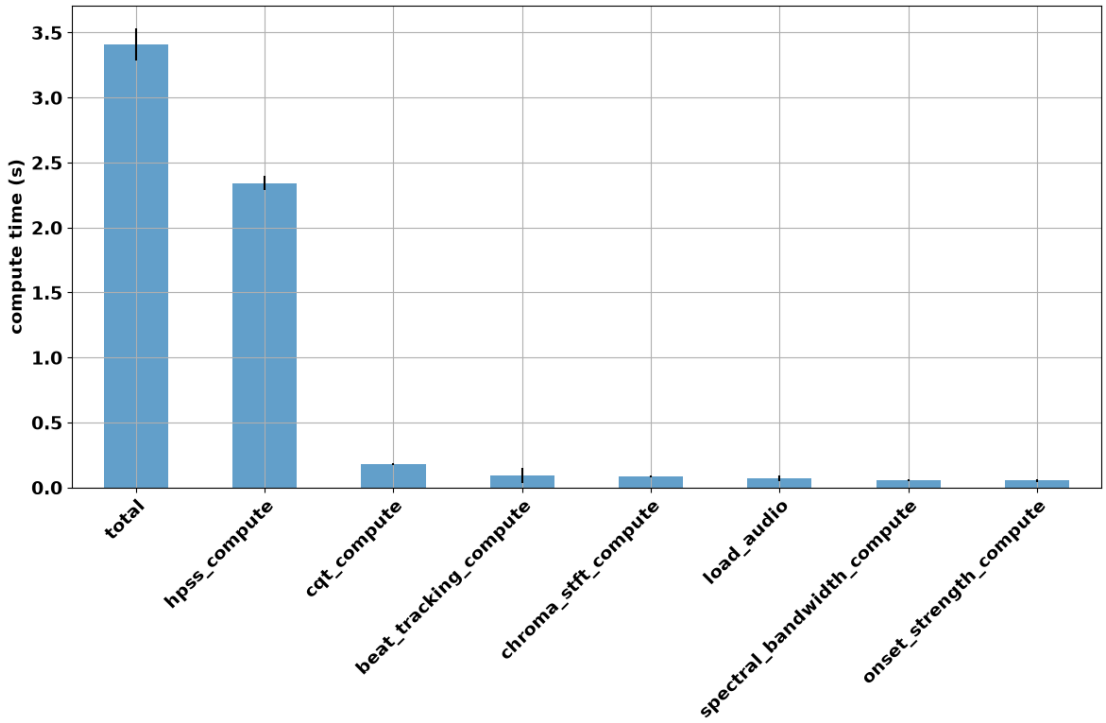


Figure 1: The computational time (in seconds) averaged on 200 audio tracks for the 7 most time consuming features. The `total` bar gives the average total computational time for one track.

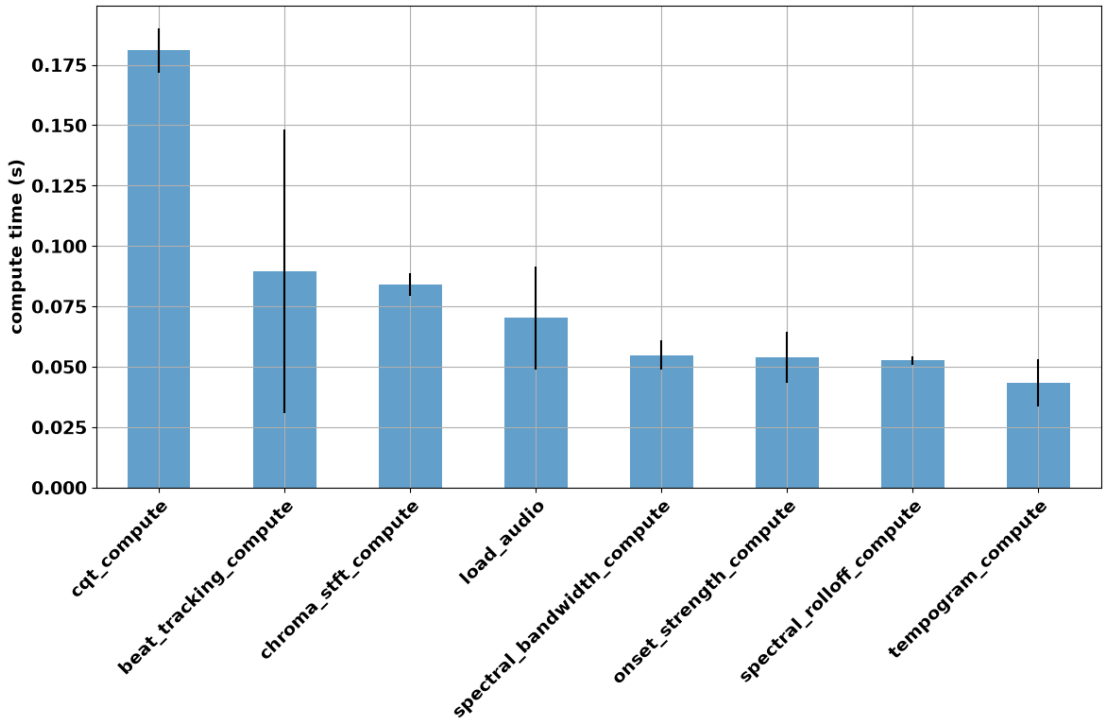


Figure 2: The computational time (in seconds) averaged on 200 audio tracks for the most time consuming features. The bars for `total` and `hpss_compute` have been removed to better see the variations on the next columns.

References

[1] Michaël Defferrard et al. "FMA: A Dataset for Music Analysis". In: *18th International Society for Music Information Retrieval Conference (ISMIR)*. 2017. arXiv: 1612.01840. URL: https://arxiv.org/abs/1612.01840.